

Adeer Khan

 adeerkhan.github.io |  adeerkhan@hotmail.com |  [adeerkhan](#) |  [Google Scholar](#)  [@AdeerKhan](#)

EDUCATION

• Korea Advanced Institute of Science and Technology (KAIST)

MSc in Civil and Environmental Engineering

◦ CGPA: 3.50/4.30 (92/100 %)

Feb 2024 - Aug 2025

Daejeon, South Korea

PROFESSIONAL EXPERIENCE

• NeoBim AI GmbH

AI Research Engineer

Oct 2023 - Present

Germany (Remote)

- Designing AI-driven architectural workflows that automate spatial planning, BIM generation, and design reasoning. designer.bluebrick.ai
- Implementing end-to-end ML pipelines connecting frontend interfaces to AWS-hosted model backends, including dataset preparation, API development, model integration, and deployment architecture.
- Evaluating 50+ generative AI repositories and fine-tuning models including Stable Diffusion, Flux, Mistral, Qwen, and LLaMA for scalable, domain-specific AEC design automation.

• Korea Advanced Institute of Science and Technology (KAIST)

Research Assistant – LLM Integrated Digital Twin Development

Feb 2024 - Oct 2025

Daejeon, South Korea

- Developed an LLM-assisted digital twin framework in Unity 3D, integrating Arduino and ESP32 IoT hardware for intelligent building system simulation and control.
- Integrated physical IoT devices with digital environments and AI-driven workflows for building monitoring, simulation, and intelligent control.

SELECTED PROJECTS

• Vitruvius – Engineering Research Agent

Open Source

[Homepage](#)

- Built an open-source agent that runs a discover–read–synthesize–verify–review loop over engineering questions, with dedicated skills for mechanical, civil, electrical, software, and architectural disciplines.
- Integrated scholarly search (OpenAlex, Semantic Scholar, arXiv) and a standards-lookup skill covering AISC, ACI, ASCE 7, IEEE, and IBC, packaged as portable Agent Skills for Command Code, OpenCode, and Pi.

• Pakistan Stock Exchange (PSX) AI Terminal

Personal Project

Supabase, Cloudflare

- Developing an AI-powered terminal for Pakistan Stock Exchange data, market analysis, and financial monitoring. Building specialized AI trading agents for market research, signal generation, and multi-agent decision support.
- Architected the full stack: Supabase-backed data layer, with the front-end and CI/CD pipeline deployed on Cloudflare for automated, production-grade releases.

SKILLS

• **AI & ML:** PyTorch, Generative AI, LLMs, Diffusion Models, Transformers, AI Agents

• **Backend & Ops:** FastAPI, Docker, AWS, Linux, Cloudflare, SQLite, CI/CD

• **CAD & BIM:** AutoCAD, SketchUp, Speckle, Revit, Rhino-Grasshopper, IFC

• **AI Development:** OpenCode, Cursor, Hermes, ComfyUI

• **Frontend:** React, Zustand, Three.js, Vite, WebGL

• **Languages:** Python, TypeScript, JavaScript, C#

• **Video & 3D:** Blender, Unity3D, CapCut

• **Data & GIS:** QGIS, pandas, NumPy, scikit-learn

• **Databases:** PostgreSQL, Supabase

COURSES & CERTIFICATIONS

[Machine Learning Specialization](#)

[Imagery, Automation, and Applications](#)

[Introduction to DevOps](#)

[Introduction to Computer Vision and Image Processing](#)

[Introduction to Software Engineering](#)

[ML Summer School 2026](#)